Installation and Usage Guide for Neo4j Database Management Application

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This guide will walk you through setting up the environment, installing necessary dependencies, running the application, and using it effectively.

0.1 Prerequisites

Before we begin, ensure you have the following installed on your system:

1. **Python:** Version 3.8 or above.
2. **Neo4j Desktop or Server:** Version 5.0 or above.
3. **Pip:** Python package installer, which typically comes with Python installations.

0.1.1 Installing Neo4j and Creating the Database

- **Download Neo4j:** You can download the Neo4j Desktop from Neo4j's official website: **https://neo4j.com/deployment-center/?desktop-gdb**. Follow the installation instructions specific to your operating system.
- **Create a new empty database and Start Neo4j-default database:** Once installed, open Neo4j Desktop, create a new local project by giving it a name and password and start the neo4j-default database.

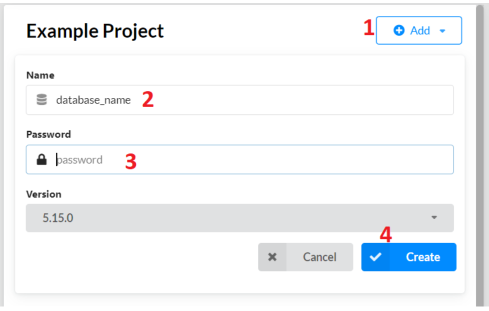

Figure 1: New Empty Database Creation

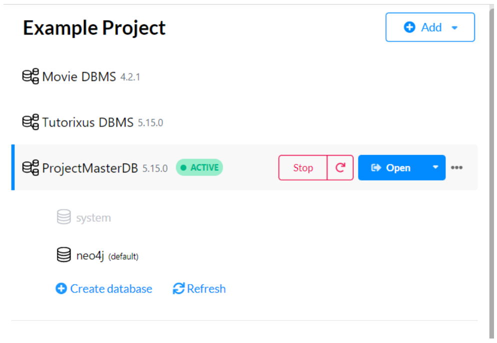

Figure 2: Starting the newly created database

0.2 Running the Application

For this project, we provided two alternatives for running the application. Either by running the **.exe** or by running it through the terminal using the **python main.py** command in the corresponding application directory.

0.2.1 Using the executable (.exe) program

This is the simplest method involving an executable program to ease work and increase user friendliness. This executable program regroups all elements to provide the required environment fur running the application

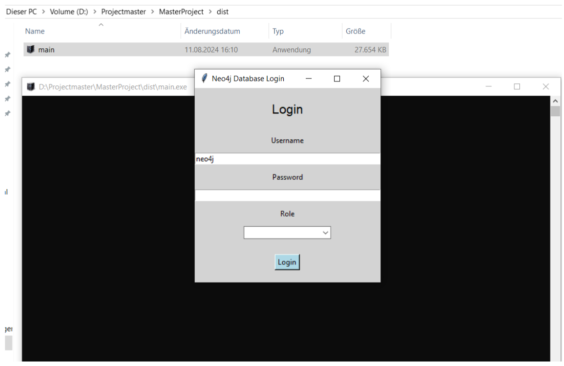

Figure 3: Running the executable program

0.2.2 Running through python command on the terminal

Setting Up the Environment

- Python Environment Setup: If you don't already have Python installed, download it from Python's official website: **https://www.python.org/downloads/**. Ensure that you add Python to your system's PATH during installation.

- Create a Virtual Environment (Optional but recommended) by running the following command: **python -m venv neo4j-env** And activate the virtual environment: **neo4j-env**

- Open the cmd-Terminal in the directory containing the main.py file. This is to make sure we are located in the correct path to run python command. To so, you can open the folder containing the main.py and press: **shift + right mouse button** and then select **Open Powershell window here** on the pannel displayed.

- In powershell, run the following command to open the terminal: **start cmd**

- Install Required Python Packages through the terminal: After setting up the virtual environment, install the required packages by running: **pip install neo4j tkinter reportlab**. These packages are: **neo4j:** To interact with the Neo4j database. **tkinter:** For creating the GUI. **reportlab:** For generating PDF reports from query results.

- Run the following command to start program: **python main.py**

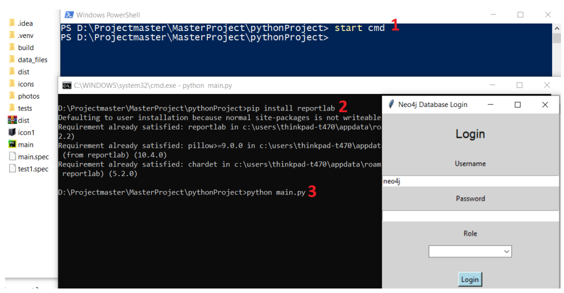

Figure 4: Running the Application through Terminal

Alternatively, open the main.py in any code editor like **Pycharm or Visual Studio Code** and in the terminal console, try install all necessary dependencies mentioned above like neo4j, tkinter and reportlab using the above commands. Once all dependencies have been install, run **python main.py** to launch the program and use.

0.3 Using the Application

0.3.1 Login Page

- **Username:** Enter the Neo4j username (default is neo4j).
- **Password:** Enter the Neo4j password.
- **Role:** Select either Admin or User. The Admin Role Grants full access to create nodes, create relationships, and drop the database. The User Role Allows the user to run queries and view results.

0.3.2 Admin Mode

Node Creation:

- Enter the node label in the provided input field.
- Click on the "Load and Create Node" button to import a CSV file that will populate the node database automatically.

Relationship Creation:

- Specify the start node label/name, relationship label, and end node label in the corresponding input field in that order.
- Make sure the two nodes required to create the relationship have already been created and exist in the neo4j database
- Click on the "Load and Create Relation" button to import a CSV file containing relationship data.

Delete Node/Relationship:

- Input the node label or relationship label to delete.
- Click the appropriate button to delete the node or relationship.

Drop Database:

- Use the "drop database" button to drop the entire database, removing all nodes and relationships.

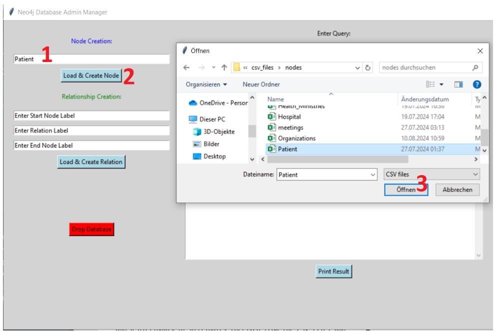

Figure 5: Node Creation Steps

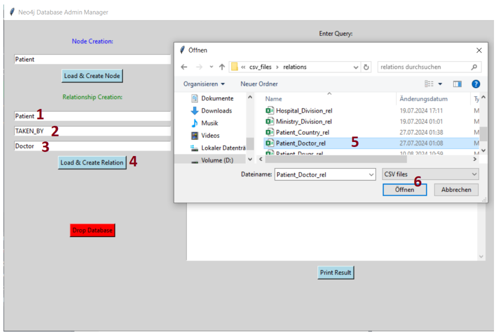

Figure 6: Relationship Creation Steps

0.3.3 User Mode

Executing Queries:

- Enter the desired query in the input field.
- Click "Execute Query" to run the query and view results.

Printing Results:

The "Print Result" button is common to both users and admin and allows users to export query results as a PDF document.

Messages

The right-hand panel provides feedback, displaying messages related to actions taken within the application (e.g., nodes created, errors, etc.).

Close Button

This button is common to both user and admin and it is used to exit the application gracefully.

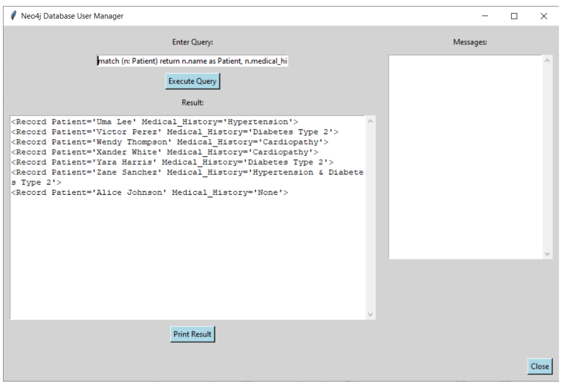

Figure 7: User Mode